

PROPOSED APPROACH TO WEBSITE DESIGN, YOUTH ENGAGEMENT, AND AI/CHATBOT FUNCTIONALITY

1. Introduction

Neptune Technology proposes a user-centred, evidence-driven, and innovation-focused approach to designing and developing African Disability Forum (ADF) next-generation digital platform. Our approach recognises that the new website is more than an online presence, it is a strategic communications and knowledge-sharing platform that must strengthen citizen engagement, amplify African Disability Forum (ADF) voice, improve access to research and evidence, and foster meaningful participation among diverse audiences, particularly young people.

The proposed solution combines modern user experience (UX) design, enterprise-grade technology, Artificial Intelligence (AI), and accessibility standards to deliver a secure, responsive, and future-ready platform. By integrating intelligent search, conversational AI, responsive interfaces, and interactive engagement features, the website will become a dynamic digital ecosystem that supports African Disability Forum (ADF) mission across East Africa while remaining easy to manage and scalable for future growth.

2. Website Design Approach

Neptune Technology will adopt a **Human-Centred Design (HCD)** methodology that places users at the centre of every design decision. The design process will begin with stakeholder consultations, user research, and an audit of the existing website to understand audience needs, information-seeking behaviour, and content priorities. Insights gathered during this phase will inform the information architecture, navigation structure, and visual identity of the new platform.

The website will be designed to accommodate the needs of African Disability Forum (ADF) diverse audiences, including citizens, researchers, journalists, donors, development partners, policymakers, and staff. Content will be organised through clear information architecture supported by intuitive navigation, allowing users to locate publications, programmes, news, events, and organisational information with minimal effort.

The visual design will reflect African Disability Forum (ADF) brand identity while presenting a modern and engaging interface that inspires confidence and encourages exploration. A consistent design system including typography, colour palette, iconography, and reusable interface components will ensure coherence across the website and simplify future enhancements.

To maximise accessibility and usability, the website will comply with **Web Content Accessibility Guidelines (WCAG) 2.2 Level AA**. Features such as keyboard navigation, alternative text for images, sufficient colour contrast, accessible forms, and semantic HTML will ensure that people with disabilities can use the platform effectively.

Recognising the widespread use of smartphones across East Africa, the website will follow a **mobile-first, responsive design** strategy. Every page and feature will adapt seamlessly to desktops, tablets, and mobile devices, ensuring consistent performance regardless of screen size or device.

Performance optimisation will be integrated into the design process through image optimisation, lazy loading, browser caching, content delivery networks (CDNs), and efficient coding practices. These measures will reduce page load times, improve user satisfaction, and enhance search engine rankings.

3. Youth Engagement Strategy

Young people represent one of African Disability Forum (ADF) most important stakeholder groups and are central to the organisation's mission of promoting citizen agency and civic participation. Neptune Technology's approach is to design a platform that is not only informative but also engaging, interactive, and relevant to digitally connected youth.

The website will feature a dedicated youth-focused experience that presents content in accessible and visually appealing formats. Long reports and policy papers can be complemented by executive summaries, infographics, short videos, podcasts, and interactive explainers that make complex issues easier to understand and share.

Interactive features such as opinion polls, surveys, discussion prompts, and event registration will encourage active participation rather than passive consumption of information. Social media integration will enable users to share stories, publications, and campaign content directly from the website, extending African Disability Forum (ADF) reach and encouraging peer-to-peer engagement.

The platform will also highlight opportunities for youth involvement, including internships, fellowships, volunteer programmes, events, and calls for participation. These opportunities will be organised within a dedicated section that allows young people to discover and respond to engagement opportunities quickly.

To foster a sense of community, the website will support storytelling through multimedia content that showcases the experiences, achievements, and voices of young people participating in African Disability Forum (ADF) programmes. This human-centred approach will make the website more relatable and strengthen emotional connections with its audiences.

Gamified engagement elements such as interactive quizzes, featured campaigns, or knowledge challenges may also be incorporated where appropriate to encourage learning and sustained interaction, particularly around governance, civic participation, and public accountability.

4. Artificial Intelligence and Chatbot Functionality

Neptune Technology proposes the implementation of an AI-powered Virtual Assistant that transforms the website into an intelligent knowledge platform. Rather than relying solely on traditional search functionality, visitors will be able to interact with the website conversationally, asking questions in natural language and receiving accurate, context-aware responses.

The chatbot will be built using a **Retrieval-Augmented Generation (RAG)** architecture. This approach enables the AI to retrieve information from African Disability Forum (ADF) own approved knowledge base including publications, reports, programme pages, policy briefs, news articles, and frequently asked questions before generating a response. By grounding answers in verified organisational

content, the chatbot will provide reliable information while reducing the risk of inaccurate or fabricated responses.

The AI Virtual Assistant will support a wide range of user needs, including:

- Locating publications, reports, and policy briefs.
- Explaining African Disability Forum (ADF) programmes and thematic areas.
- Answering frequently asked questions.
- Guiding visitors to relevant pages and resources.
- Providing contact information and office locations.
- Directing users to partnership, funding, or career opportunities.
- Assisting users in navigating the website.

The chatbot will be available across desktop and mobile devices, ensuring that users can access support regardless of how they visit the website. Where appropriate, multilingual capability can be incorporated to support English and Kiswahili content, improving accessibility for regional audiences.

To ensure responsible AI deployment, Neptune Technology will implement governance measures that include human oversight of the knowledge base, transparent disclosure that users are interacting with an AI assistant, administrative controls for content updates, usage monitoring, and privacy safeguards. These measures will ensure that AI enhances user experience while maintaining trust, accountability, and compliance with data protection requirements.

5. Delivering Long-Term Value

Neptune Technology's proposed approach goes beyond the delivery of a modern website. It establishes a sustainable digital platform that supports African Disability Forum (ADF) strategic objectives by combining user-centred design, meaningful youth engagement, and intelligent technology.

The resulting platform will:

- Provide an accessible, secure, and responsive digital experience for all users.
- Increase the visibility and discoverability of African Disability Forum (ADF) research and programme outputs.
- Encourage greater participation among young people through interactive and multimedia content.
- Improve access to information through AI-assisted knowledge discovery and conversational support.
- Enable Twaweza staff to manage and update content efficiently using an enterprise content management system.
- Provide analytics and insights that support evidence-based communications and continuous improvement.

Through this integrated approach, Neptune Technology will deliver a future-ready website that strengthens African Disability Forum (ADF) digital presence, deepens stakeholder engagement, and positions the organisation as a leader in innovative, citizen-focused communication across East Africa.